
When recovery does not validate a reporting rule: a synthetic trial example and an empirical audit

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Abstract

1 Recovering a reference effect does not establish that a simulator supports the
2 intended decision. We examine this gap through a synthetic external-control
3 example and an empirical self-audit. In the former, using a reference for the
4 wrong population favors an estimator of the wrong effect; an enrolled-population
5 reference reverses that ranking. In the latter, we audit a 50-cell reporting rule in
6 two colorectal single-cell cohorts that had been justified by point-estimate recovery.
7 In 5,000 simulations per setting, its nominal 95% percentile intervals reject a true
8 null in 9.8–32.2% of studies at the reporting boundary. A paired comparison with
9 Welch intervals improves effect coverage but leaves null rejection at 8.9–30.8%;
10 substituting a Welch interval does not repair the rule. An exact variance calculation
11 shows that, under a halving of expression, the sampled reference removes three
12 quarters of the reference-arm variance contribution from the error being assessed;
13 at 800 cells, apparent coverage exceeds fixed-pool coverage by 5.8–6.8 percentage
14 points. We also document grid-dependent threshold selection. The practical lesson
15 is to specify the target, test null and alternative performance together, and evaluate
16 an interval and its reporting threshold as a combined rule.

17 1 Motivation: validate the decision, not just recovery

18 A clinical simulator may reproduce a treatment-effect reference accurately while providing inadequate
19 evidence for the decision it is meant to support. The reference may average effects over the wrong
20 population; even with the right target, recovering a point estimate does not establish the coverage
21 or false-positive rate of the reported interval. Our thesis is that validation must address the target,
22 interval and reporting policy together. We examine this gap through a constructed external-control
23 example and a documented single-cell self-audit, with different evidential roles.

24 The external-control example isolates a target-population error relevant to single-arm trial simulation
25 (Section 2). A pooled-population estimator appears exact against its observed-data reference yet
26 overstates the benefit among enrolled records. Changing the reference to their known potential-
27 outcome effect reverses the estimator ranking. This establishes a possible failure and a diagnostic
28 control in a fully specified synthetic setting.

29 The empirical audit examines an actual validation claim in our unpublished colorectal single-cell
30 project (Section 3). Point-estimate recovery had justified a 50-cell reporting boundary. Direct
31 evaluation finds excessive false positives with both percentile and Welch intervals, and changing the
32 scoring target exposes misleading apparent coverage. Controlled grid comparisons document how the
33 selected count depends on grid and binning choices. Together, these analyses trace how a reassuring
34 recovery result can fail to justify the rule used downstream.

Contribution and prior work. The contribution is a reproducible worked audit, not a new validation principle. Simulation-study methodology already separates estimands, methods and performance measures [Burton et al., 2006, Morris et al., 2019]; ICH E9(R1) requires the clinical question to identify its target population [International Council for Harmonisation, 2019]. Generator–analysis alignment is studied through congeniality [Meng, 1994], plasmode and semi-synthetic benchmarks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019] and the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]; generators can also disadvantage good estimators [Shaw et al., 2026]. Clinical external-control validation already uses held-out studies to assess bias and type-I error [Ventz et al., 2019]. Prognostic adjustment retains trial randomization [Schuler et al., 2022], and target-trial emulation makes the causal question and design explicit [Hernán and Robins, 2016]. We apply these established distinctions to an inspectable reporting-rule decision and a controlled population mismatch. Simulation-based calibration (SBC) checks Bayesian computation using ranks under prior-predictive simulation [Talts et al., 2018]; its sensitivity depends on the test quantities [Modrák et al., 2025]. Our frequentist audit instead measures coverage and rejection at specified effects and empirical source pools, documenting where recovery was used to justify a reporting rule. A supporting learned-generator check isolates why increasing generator complexity does not break an estimator–reference identity (Appendix C). Section 4 draws out the resulting checks for clinical simulation; Appendix F provides further context.

2 Clinical-simulation example: validating the wrong population

Consider validating an estimator for a single-arm trial using an external control arm (ECA). This constructed experiment isolates a mismatch between the population named by the clinical question and the population used in validation. The estimator and its observed-data reference deliberately share the same population mistake; exact recovery may itself alert an analyst. This is a pedagogical stress test, not a claim that the mistake is common. A fixed synthetic registry supplies a single-arm trial and three external controls per enrolled patient. Enrollment favors heavily pretreated strata; induced benefits vary by stratum and latent individual responsiveness. The intended target is the finite-cohort mean treatment effect among enrolled records $\bar{E}$, available to the simulator through both potential outcomes:

$$\tau_E = \frac{1}{|\bar{E}|} \sum_{i \in \bar{E}} \{Y_i(1) - Y_i(0)\}, \quad \hat{\tau}(w) = \sum_g w_g (\bar{Y}_{1g} - \bar{Y}_{0g}). \quad (1)$$

Here g indexes treatment-history strata, and the observed means use treated enrollees and untreated external controls. Enrolled standardization uses $w_g = n_{Eg}/|\bar{E}|$; pooled standardization uses $w_g = (n_{Eg} + n_{Cg})/(|\bar{E}| + |\bar{C}|)$ for controls $\bar{C}$. Both use the same stratum contrasts. Because effects vary across strata and enrollment changes their mix, these weights answer different questions. Pooled standardization is inappropriate for the stated enrolled target, not intrinsically invalid.

We evaluate 200 studies at each of six trial sizes from 50 to 1,600, holding the synthetic registry fixed. At the largest size, Table 1 compares the same estimates against three references: the observed pooled contrast, the pooled potential-outcome effect, and τ_E . Appendix D specifies the generator and denominators.

Table 1: Reference choice reverses the estimator ranking in the synthetic external-control example. Bias and RMSE use 200 studies with 1,600 enrolled records and 4,800 controls. Columns identify the population used to average observed stratum contrasts; only enrolled standardization matches the intended target. All entries are in abstract outcome units.

Validation reference	Pooled standardization		Enrolled standardization	
	Bias	RMSE	Bias	RMSE
Observed contrast, pooled	0.000	0.000	-0.872	0.876
Potential outcomes, pooled	+0.003	0.115	-0.869	0.875
Potential outcomes, enrolled	+0.885	0.892	+0.013	0.105

At 1,600 enrolled patients, pooled standardization has a median estimate of 1.890 outcome units while the enrolled records’ median true effect is 1.003. These medians fall on opposite sides of the prespecified illustrative threshold of 1.5; this is not an estimated probability of a wrong decision.

Against the observed pooled contrast, the pooled estimator has zero error by construction, while enrolled standardization appears inaccurate (Table 1). Using the enrolled potential-outcome reference reverses that ranking. Merely using potential outcomes is insufficient: averaging them over the pooled population still favors the estimator of the wrong target. The target must specify both the effect and the population over which it is averaged.

Controls distinguish agreement from target alignment. At zero case-mix shift, pooled standardization still reproduces its observed-data reference but its bias against the enrolled effect is only -0.002 units. Equality therefore does not establish that an estimate is wrong. Under the full shift, pooled weighting balances the measured strata between arms to numerical precision while their weighted distribution differs from the enrolled distribution (maximum standardized difference 0.964). Between-arm balance does not establish target-population alignment. At the smallest and largest trial sizes, root mean squared error against τ_E is respectively 0.608 and 0.105 for enrolled standardization, and 1.082 and 0.892 for pooled standardization. Increasing sample size therefore does not remove the population mismatch in this experiment.

For clinical-simulation validation, the actionable check is to compare the analysis population with the population named by the estimand, in addition to checking between-arm balance. Potential-outcome access solves the missing counterfactual problem inside this simulation; it does not choose the target population. The next section examines the complementary problem: even a correctly specified target requires direct evaluation of interval performance and the reporting boundary.

3 Empirical audit: when recovery fails to validate a reporting rule

Application and reporting policy. The empirical application separates two explanations for differentiation-marker loss in colorectal cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that remain (an *intrinsic* change). These explanations suggest different biological interpretations. An empirical comparison of per-cell means requires the relevant population to be observed in both tissues. Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their mean per-cell output. A reference-based decomposition, related to demographic standardization and Oaxaca–Blinder decompositions [Kitagawa, 1955, Oaxaca, 1973, Blinder, 1973], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (2)$$

The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells, the corresponding conditional mean itself has no referent. Substituting zero in either case turns an unavailable contrast into an apparent effect.

The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n < 50$, and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue. Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations examined below also retain raw arithmetic before this reporting gate. We evaluate the boundary directly and retain pre-gate intervals at smaller counts to diagnose the curve’s behavior.

The claim being re-evaluated. An internal assessment (Artifact A) treated the 50-cell boundary as validated because recovery ratios were 0.85–1.08 above it and 1.18 in the 20–49-cell band, using six draws per setting at 1,200 cells per arm. The design record (Artifact B) instead required at least 90% interval coverage and 80% exclusion of zero at a halving of expression. Appendix F and the anonymized supporting record make this mismatch inspectable.

3.1 Simulation and inferential targets

The design document fixed a detectable halving of per-cell output ($s = 0.5$), coverage of the target in at least 90% of defined outputs, and discrimination of at least 80%, defined as the probability that the interval excludes zero at that effect. The implementation scored coverage against a drawn-reference benchmark; we retain that score and add coverage against the generating pool mean. The candidate *ok* is the first evaluated count or bin meeting both criteria; *wide* is the first meeting coverage alone.

123 The committed values are 50 and 20. The first-crossing definition does not require larger counts to
 124 qualify.

125 The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from
 126 GSE132465 (SMC) and six from GSE144735 (KUL3). We analyze GUCA2A in the differentiated
 127 population under fixed lineage labeling. Each draw samples, with replacement, 2,000 cells per arm
 128 from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool
 129 combines cells from both tissues before drawing either synthetic arm. Because the design document
 130 did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Both arms sample the
 131 same eligible pool, with an imposed expression shift. A nominal 95% percentile interval resamples
 132 mature cells in both arms, with fractions held fixed and 200 inner draws unless stated otherwise.
 133 Undefined intervals are excluded from rate denominators and reported as abstentions (Appendix B).

134 The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (3)$$

135 where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Let P be the eligible two-patient
 136 source pool, M_P its mature-cell count, and Y_j its unshifted target-gene counts. The population
 137 parameter of this finite empirical distribution is instead

$$\mu_P = \frac{1}{M_P} \sum_{j \in P, \text{mature}} Y_j, \quad \theta_P(s) = f_N(s - 1)\mu_P. \quad (4)$$

138 We compute μ_P from all eligible source cells before either synthetic arm is sampled. Given P , the
 139 generator has $E[\bar{Y}_N] = \mu_P$ and $E[\bar{Y}_T] = s\mu_P$, making Equation 4 its true intrinsic mean contrast.
 140 It is fixed across cell draws from that pool. Patient-pair selection changes P ; reported coverage
 141 averages this conditional-pool question over the selected pairs. An all-patient pooled mean would use
 142 a different population from the generator’s two-patient draw. Neither target supplies inference over
 143 new patients.

144 Two complementary runs separate historical selection from direct evaluation. The controlled eight-
 145 seed grid retains inclusion of Equation 3. An initial 200-replicate diagnostic reproduced its first-seed
 146 results and added fixed-pool coverage and null rejection. The precision follow-up uses 5,000 fresh
 147 studies per cohort, pool, count (5, 50, 100, 800) and shift ($s = 0.5, 1$), split across five seed streams:
 148 160,000 studies in total. It samples the exact marginal law of mature-cell counts in the original
 149 generator; non-mature cells do not enter this contrast. On each study we compare the original
 150 200-resample percentile interval with a Welch–Satterthwaite interval [NIST/SEMATECH]. This
 151 standard unequal-variance t approximation is a comparator, not an assumed solution for sparse counts.
 152 All 320,000 intervals are finite. The largest binomial Monte Carlo standard error is 0.71 percentage
 153 points, conditional on the empirical source pools. Appendix A specifies the sampling and interval
 154 calculations.

155 3.2 Fixed-pool coverage and false positives

156 The larger run resolves the 50-cell coverage comparison (Table 2). Percentile coverage is below
 157 the recorded 90% criterion in three conditions, whose Wilson upper endpoints are below 90%.
 158 SMC/reference has 91.0% coverage, with its lower endpoint above 90%. Thus the rule does not
 159 fail coverage uniformly. It does fail the combined assessment: null rejection is 9.8–32.2%, above
 160 the nominal 5% in every condition, and detection is 47.5–76.2%, below the required 80%. All four
 161 detection upper endpoints are below 80%; all four null-rejection lower endpoints exceed 5%. These
 162 intervals measure simulation precision, not uncertainty over additional patients.

163 3.3 Shared-reference scoring removes part of the estimation error

164 At 800 cells, evaluating the percentile intervals against the sampled reference gives 98.9–99.6%
 165 inclusion, compared with 92.3–93.6% coverage of the fixed-pool effect. The paired differences are
 166 5.8–6.8 percentage points with Monte Carlo standard errors of 0.33–0.36 points. Only the scoring
 167 target changes. At five cells the direction reverses; shared randomness need not inflate inclusion at
 168 every sample size.

169 The increasing discrepancy at large tumor-arm counts has an analytic explanation. The estimate is
 170 $\hat{\theta} = f_N(\bar{Y}_T - \bar{Y}_N)$, whereas its error against the drawn benchmark is $\hat{\theta} - i_{\text{req}} = f_N(\bar{Y}_T - s\bar{Y}_N)$.

Table 2: At the 50-cell reporting boundary, substituting Welch intervals improves coverage but does not restore the required performance. Rates (%) and 95% Wilson Monte Carlo intervals use 5,000 studies per cohort, pool and effect. Methods share the same samples. Coverage targets the fixed-pool effect at $s = 0.5$; null rejection uses $s = 1$; detection uses $s = 0.5$. Targets are at least 90% coverage, nominal 5% null rejection and at least 80% detection.

Cohort / pool	Interval	Coverage	Null rejection	Detection
SMC / pooled	Percentile	87.1 [86.2, 88.0]	25.4 [24.2, 26.6]	51.1 [49.7, 52.4]
SMC / pooled	Welch	90.2 [89.3, 91.0]	23.5 [22.3, 24.7]	50.7 [49.3, 52.1]
SMC / reference	Percentile	91.0 [90.2, 91.8]	9.8 [9.0, 10.7]	76.2 [75.0, 77.3]
SMC / reference	Welch	92.5 [91.8, 93.2]	8.9 [8.1, 9.7]	77.6 [76.5, 78.8]
KUL3 / pooled	Percentile	84.7 [83.6, 85.6]	32.2 [31.0, 33.5]	49.0 [47.6, 50.4]
KUL3 / pooled	Welch	89.7 [88.8, 90.5]	30.8 [29.5, 32.1]	49.0 [47.6, 50.3]
KUL3 / reference	Percentile	86.3 [85.3, 87.2]	19.0 [18.0, 20.1]	47.5 [46.1, 48.8]
KUL3 / reference	Welch	88.1 [87.2, 89.0]	18.8 [17.7, 19.9]	48.2 [46.8, 49.6]

For independent arm draws conditional on P , their error variances differ by

$$\text{Var}(\hat{\theta} - \theta_P \mid P) - \text{Var}(\hat{\theta} - i_{\text{req}} \mid P) = f_N^2(1 - s^2) \frac{\sigma_P^2}{n_N}, \quad (5)$$

where σ_P^2 is the source-pool expression variance and $n_N = 800$. At $s = 0.5$, the drawn benchmark removes three quarters of the reference-arm variance contribution from the error being assessed, while the bootstrap continues resampling both arms. This explains why near-perfect benchmark inclusion can accompany lower coverage of the generating parameter. Skewed finite-sample errors still determine the actual percentile-interval rates.

The new runs assess four specified counts, rather than selecting a replacement threshold. The eight-seed grid below remains scored against its original benchmark and cannot be reinterpreted as fixed-pool calibration.

3.4 Welch intervals improve coverage but do not repair the rule

Welch intervals increase fixed-pool coverage at 50 cells by 1.5–5.0 percentage points on the same draws (paired Monte Carlo standard errors 0.23–0.32 points). They also reduce null rejection, but it remains 8.9–30.8% and every lower confidence endpoint exceeds 5%. Welch intervals have no bootstrap resampling noise, so their excess null rejection on the same draws shows that finite inner-bootstrap noise cannot be the sole explanation for the failure shared by both methods. Detection remains below 80% even at the upper confidence endpoints. Substituting Welch intervals therefore fails the joint requirements at 50 cells. The follow-up holds the percentile budget at 200; it does not measure how a larger budget would change those intervals.

Bias and skewness corrections, including bias-corrected and accelerated (BCa) and studentized bootstrap intervals, are established in the bootstrap accuracy literature [DiCiccio and Efron, 1996]. This experiment quantifies the operating characteristics of the implemented rule and one comparator; it does not discover the general vulnerability of simple intervals to skewed sampling distributions.

Why detection can fall as cell count rises. The apparent detection curve combines effect sensitivity with a changing false-positive rate. Figure 1 reproduces the pooled-draw trough with greater precision. For KUL3, percentile detection falls from 49.0% at 50 cells to 44.0% at 100, then rises to 57.7% at 800; null rejection falls from 32.2% to 20.3% to 6.5%. At 50 and 100 cells, the median sample-based standard error is respectively 0.66 and 0.86 times the exact pool-conditional standard deviation. The fraction of draws with that ratio below one half falls from 29.1% to 7.6%. These diagnostics support a loss of spuriously confident rejections as the sampled variance becomes better represented, before increasing signal-to-noise dominates. All-zero tumor samples occur in only 0.32% of the 50-cell draws and none at 100: the trough is not explained by all-zero samples alone.

The five-cell diagnostic makes the problem especially clear: pooled percentile null rejection is 66.1% in SMC and 71.9% in KUL3. Under the halving, 46.2% and 41.9% of tumor samples are all zero. Resampling cannot recover an unobserved positive tail, although the reference arm still contributes uncertainty. The original reporting gate already abstains here; the 50-cell failures concern returned answers directly.

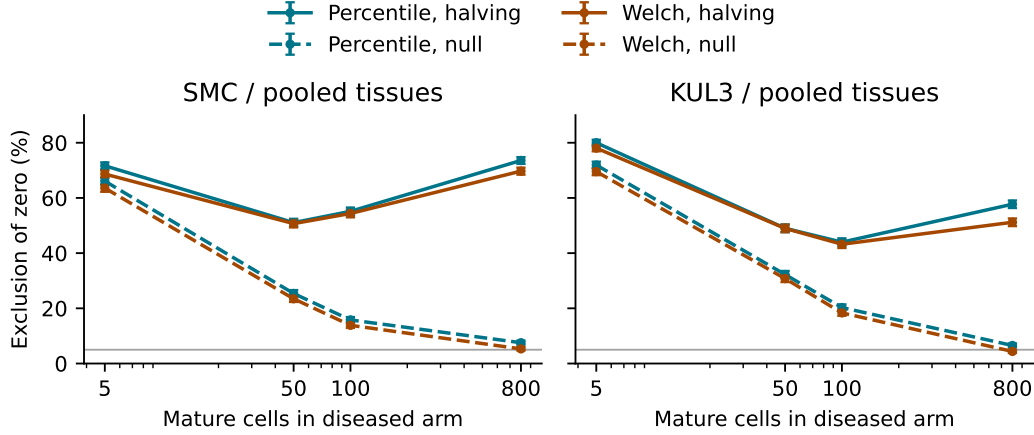


Figure 1: Detection must be read alongside null rejection. Pooled-tissue simulations use 5,000 studies per point; bars are 95% Wilson Monte Carlo intervals. Solid curves use a halving of expression and dashed curves the null. The horizontal line marks nominal 5% rejection. Five-cell intervals are diagnostic calculations before the reporting gate, which abstains below 20. Both methods use the same samples; their apparent sensitivity at sparse counts accompanies excessive false positives.

3.5 Count grids change candidate cutpoints

Historical grids returned reference-only binned candidates of 100, 90 and 70, but changed random streams, bin boundaries and replicate budgets together (Appendix B). To isolate these choices, we simulated the union of counts once per cohort, pool and each of eight seeds, with 200 outer replicates and 200 inner bootstrap draws. Every grid summary reuses the same outputs at shared counts. A *grid* is the set of mature-cell counts a sweep can evaluate, fixed by which mature-cell fractions it varies. The committed grid is the one recorded in the original design and reaches {0, 20, 40, 100, 200, 400, 800}; the filled and dense grids add support between and below those counts, and differ from it in nothing else. Fixed bins share boundaries across grids; adaptive bins recalculate them. Binned candidates are median counts within the selected bin, not estimated crossing locations.

With common fixed bins, SMC returns 100 on the committed grid and 70 on both expanded grids in all eight seeds. Adaptive bins return 65 on seven filled-grid seeds and 90 on one, while the dense grid returns 70 throughout. KUL3 instead returns 400 on six seeds and 800 on two, with all grid and binning schemes agreeing within each seed. Pooled draws yield no joint candidate in either cohort. The largest binomial Monte Carlo standard error in this grid is 0.035.

Table 3: Joint coverage-and-discrimination candidates from the controlled union-grid run. Entries give candidates with seed frequencies in parentheses where they vary; otherwise all eight seeds agree. Every grid summary reuses the same draws at shared counts. “None” means no tested count or bin qualifies.

cohort / pool	grid	fixed bins	adaptive bins	per count
SMC / reference	committed	100	100	100
	filled	70	65 (7), 90 (1)	50–80
	dense	70	70	45–80
KUL3 / reference	all three	400 (6), 800 (2)	same	same
Both / pooled	all three	None	None	None

At the committed count of 50, SMC reference-only discrimination ranges from 75.5% to 81.0% across seeds, with both criteria met on two of eight seeds. KUL3 reference-only discrimination is only 45.0–55.0% at that count. These are noisy selections: three SMC/reference seeds cross the criterion and fall below it at a later count. With only 200 replicates per setting, these local crossings do not establish non-monotonicity of the underlying performance curve. The 5,000-study follow-up does not evaluate counts 45, 60 or 70 and cannot resolve those crossings.

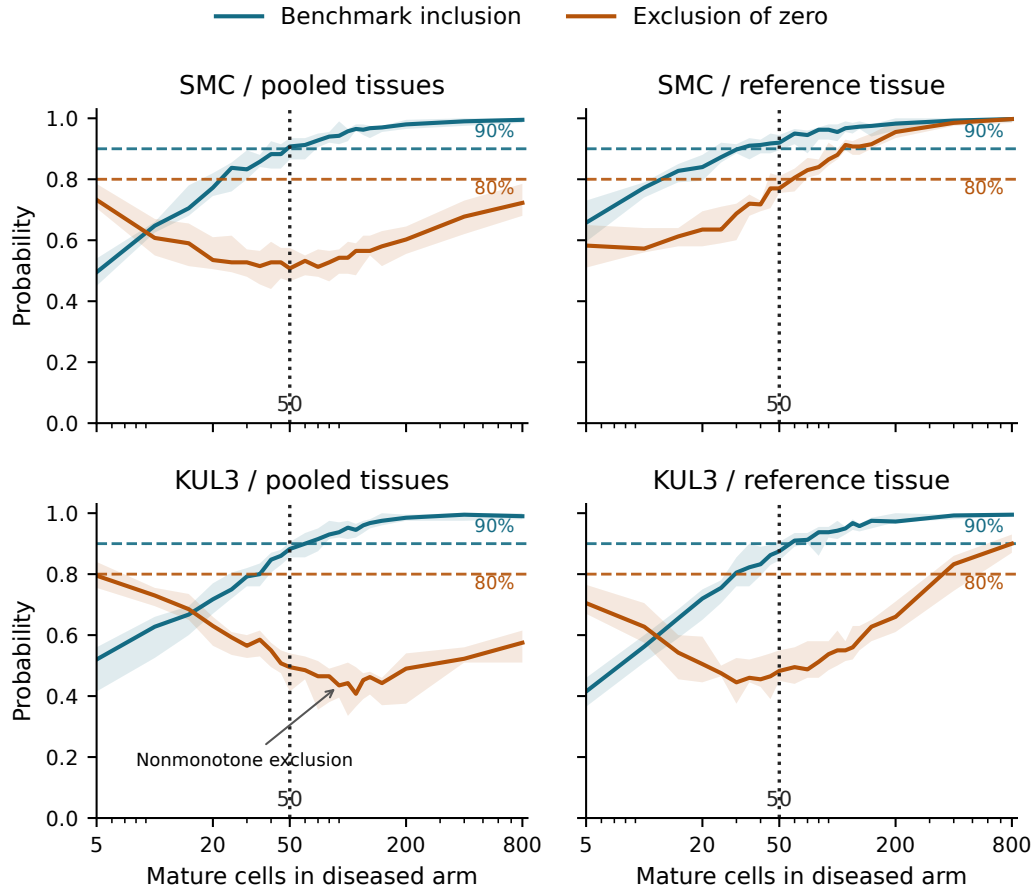


Figure 2: Historical selection diagnostic on the controlled union of counts. Benchmark inclusion is not fixed-pool coverage; Figure 1 and Table 2 provide the direct interval assessment. Rows show cohorts; columns show draw pools. Curves are median rates over eight seeds and bands span their minima and maxima, conditional on the observed cohorts. Dashed lines mark the 90% benchmark-inclusion and 80% discrimination targets; the vertical line marks the committed count of 50. Rates condition on defined intervals and coverage uses the draw-dependent reference in Equation 3. Zero-cell settings have no defined interval and are omitted. No pooled-cell candidate qualifies. Per-count reference-only candidates vary by seed and cohort (Table 3).

The per-count summaries in Table 3 supply a useful check on the bin medians. In SMC, the filled grid's first qualifying actual count is 50 on two seeds, 60 on five and 80 on one; the dense grid adds a candidate of 45 on one seed. Thus a stable binned candidate of 70 is not evidence that 70 is a uniquely determined biological threshold. A bin pools outcomes over several count settings, whereas a per-count rate addresses only that setting. Fixed bins hold the boundaries constant, but adding settings can still change both a bin's median count and its aggregate performance.

3.6 Patient influence and the cohort gap

Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort's candidate at 70, but moves the second's 400 to 300 and back to 400. Omitting one source patient changes the joint coverage-and-discrimination candidate for none of ten patients in the first cohort and three of six in the second (KUL01, KUL30 and KUL31). For the coverage-only candidate, the corresponding counts are five of ten and zero of six. These checks quantify influence within each observed cohort. Appendix B gives the biological comparability limits.

The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves the joint candidate from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts, 40 and 200. Their stability cannot locate a lower crossing outside those windows. Overlapping pairs do not add independent patients.

Expression dispersion partly explains the cohort gap. Holding tissue eligibility fixed at reference-only, the complete SMC and KUL3 mature-cell pools contain 662 and 844 cells. Across their 45 and 15 eligible patient pairs, median source-pool squared coefficients of variation ($c_P^2 = \sigma_P^2/\mu_P^2$) are 4.35 and 16.99; median zero-expression fractions are 39.3% and 62.1%. The source means are 19.5 and 11.9 raw counts, respectively (medians over pools). The KUL3 contrast therefore has substantially more relative sampling noise despite its larger source-cell pool.

For the implemented binomial thinning, the tumor-cell variance is $s^2\sigma_P^2 + s(1-s)\mu_P$. At $s = 0.5$, the exact conditional variance of the contrast relative to its squared signal is

$$\frac{\text{Var}(\hat{\theta} | P)}{\theta_P(0.5)^2} = \frac{c_P^2 + 1/\mu_P}{n_T} + \frac{4c_P^2}{n_N}. \quad (6)$$

With $n_T = 50$ and $n_N = 800$, the median signal-to-standard-deviation ratios are 3.02 in SMC and 1.53 in KUL3. This provides a distributional explanation for weaker KUL3 discrimination. It does not reproduce the exact 70-versus-400 candidate ratio: percentile-bootstrap behavior, pair heterogeneity and binning also affect that selection. The biological causes of the source-distribution difference remain unexplained.

Biological interpretation is a separate requirement. Interval calibration cannot establish a biological mechanism. The assumed retained-gene control, MS4A12, instead falls by -0.971 to -0.951 in non-sparse primary-cohort strata. Its depletion persists under broader cell inclusion and depth matching and appears in a separate bulk dataset. These checks undermine the panel's retention assumption; they do not separate composition from within-cell change. Appendix E reports this control-panel audit as a limitation of the biological application.

4 Implications for clinical-simulation validation

The two examples expose different breaks between recovery and a downstream claim. The external-control example changes the target population; the single-cell audit changes the scoring target and evaluates the reporting rule. They motivate three checks using established simulation-study practice.

State the target before constructing its reference. Distinguish a population treatment effect, a latent finite-cohort effect and an observed-data contrast. Name the population and the averaging weights. A reference need not be independent of the simulated dataset: τ_E is necessarily a property of the enrolled records. The requirement is that it represents the intended target, rather than merely reproducing the estimator.

Evaluate the interval and reporting policy jointly. Assess the whole decision rule using the interval that would actually be reported. Record bias and error against the chosen target, null rejection, detection at the specified effect, and abstention frequency. Report both conditional interval performance among returned answers and how often a valid answer is returned. A successful effect-detection rate can accompany excessive null rejection, so the former alone cannot validate an interval. The null target here is 5% because the intervals are nominally 95%.

Separate rule selection from assessment and transfer. Separate selecting a rule from assessing it. If a scalar threshold is intended to guarantee performance above that threshold, qualification must address the evaluated upper range rather than only its first passing point. Reserve independent simulation draws for the final assessment, report rate uncertainty and the tested range, and distinguish this from transfer to new patients or cohorts. Increasing resamples from an empirical pool improves

289 conditional numerical precision; it does not broaden that pool’s biological support. For an external-
290 control application, the additional question is comparability across data sources, which requires direct
291 study-level validation such as the held-out comparisons of Ventz et al. [2019].

292 5 Conclusion

293 The synthetic external-control example shows how a validation reference can favor an estimator
294 of the wrong population effect. The audited 50-cell rule fails joint interval evaluation despite
295 reassuring point-estimate recovery. A larger simulation budget distinguishes three coverage failures
296 from one passing condition, while excess null rejection and insufficient detection occur in all four.
297 Welch intervals improve coverage but do not repair the reporting boundary. In the higher-precision
298 pooled-draw study, the detection trough becomes interpretable alongside the changing false-positive
299 rate.

300 A reporting threshold must be assessed with its actual interval, explicit target and null and alternative
301 operating characteristics. The target must name the population over which effects are averaged, and
302 qualification must address the range over which the rule will report. Recovery alone supplies neither
303 guarantee.

304 6 Responsible use, limitations and impact

305 **Misuse and limitations.** NOT ESTIMABLE must not be read as “no effect” or as evidence that
306 surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42
307 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B).
308 The evidence consists of one empirical self-audit and a constructed external-control example; neither
309 validates a patient world model or a replacement reporting rule. BCa, bootstrap-*t* (studentized), and
310 Poisson- or negative-binomial-based intervals were not evaluated. Calibration covers one gene, one
311 labeling axis and one weighting on two small cohorts. It challenges this rule’s justification, not
312 cell-count rules in general. The fixed-pool analysis estimates coverage conditional on empirical
313 cell distributions; uncertainty over new patients remains unmeasured. The draw-pool choice was
314 underspecified, repeated simulations do not add independent patients, and the leave-one-patient-out
315 sensitivity remains conditional on these cohorts; patient-level population uncertainty in the candidate
316 cutpoints is unquantified. The absence-defined labeling axis remains sensitive to sequencing depth
317 (Appendix B). The external search found no affirmative case, so the documented failure remains a
318 self-audit rather than evidence about its prevalence. The external-control example uses a synthetic
319 registry and an illustrative decision threshold; it does not establish clinical benefit or support a
320 treatment decision.

321 **Data and impact.** The single-cell cohorts are public, de-identified and previously published. The
322 release policy covers derived summaries and code, rather than cell-level matrices. A system reporting
323 group-specific quantities should distinguish an unavailable comparison from an estimated null and
324 should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration
325 attempts helps prevent unsupported thresholds from being treated as validated.

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A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; stem_pole axis, lineage resolution, differentiated mature label; reference (normal) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. The historical inclusion score uses the drawn reference in Equation 3; the paired follow-up also evaluates fixed-pool coverage using Equation 4.

Historical grid budgets are reported in Appendix B; Section 3 specifies the controlled grid; the larger interval comparison is detailed next. The source cells are observed data; the external-control outcomes are simulated.

Precision follow-up and comparator. Five fresh seed streams (202609160–202609164), each with 1,000 studies per setting, provide 5,000 draws at each of four counts and two shifts in four cohort/pool combinations. Every draw uniformly chooses a two-patient pair, samples 800 reference and n_T tumor mature-cell counts independently with replacement, and thins tumor counts binomially. This is the original generator’s marginal law for this contrast, not a replay of its random stream. All 120 source-pair summaries match the earlier analysis. The vectorized percentile routine agrees with the original implementation on 32 actual-pool checks; the Welch implementation agrees with an independent library routine.

Writing $v_a = S_a^2/n_a$ for arm a , the Welch interval is $\hat{\theta} \pm t_{0.975, \nu} f_N \sqrt{v_N + v_T}$, where $\nu = (v_N + v_T)^2 / [v_N^2/(n_N - 1) + v_T^2/(n_T - 1)]$. Zero estimated total variance would give an unavailable comparator interval; none occurred. For the sparsity diagnostic, sample-based standard error is compared with $f_N \sqrt{\sigma_P^2/n_N + (s^2\sigma_P^2 + s(1-s)\mu_P)/n_T}$. Aggregate outputs retain signed rejections, widths, seed-specific rates and paired differences. The settings and single comparator were fixed before this follow-up was run; it remains an exploratory study on previously examined source data. The earlier 200-draw target diagnostic is retained in the supporting summaries.

B Calibration details and limits

Historical grids. Before the controlled run, SMC reference-only binned candidates were 100, 90 and 70 on the committed, filled and dense grids, with 500, 500 and 200 replicates per setting. These runs changed random streams and bin boundaries as well as evaluated counts; Section 3 separates those choices. KUL3 reference-only candidates were 400 or 800, and no pooled-draw candidate qualified. Its largest setting sampled 800 mature cells per arm with replacement from 844 distinct source cells, not additional patients.

Recovery accounting and unavailable contrasts. The four diagnostic sweeps comprise 1,711,200 replicates: 1,283,400 with nonzero requested intrinsic reference and 427,800 with zero requested reference, for which ratios are undefined. The first 50-replicate sweep has 119,600 draws, partitioned into 89,700 and 29,900, respectively. The maximum estimate–realized-reference residual is zero throughout. At a zero-cell setting the raw ratio is $1/(1-s)$ when the denominator is nonzero. These raw diagnostic values arise from the harness setting an empty-arm mean to zero; they do not define an effect in absent cells. The reporting gate abstains and the interval is undefined. Separately, undefined intervals were previously counted as failures through comparisons with NaN; corrected rates exclude them and report abstentions. An all-abstaining bin has no coverage estimate. One historical seed gives $ok = wide = 45$, showing that a three-state specification need not yield three distinct fitted regions.

Biological scope and patient uncertainty. The finest annotation resolution yields abstention in 28/28 patients under one population definition and 24/28 under the other; coarser definitions do

not resolve identifiability. The coarsest definition labels every epithelial cell mature and forces the compositional term to zero. In a separate comparability analysis over three cohorts and 42 patients, 83 of 84 housekeeping-detection control rows pass a 0.10 tolerance despite expression changes of 1.19–1.64 fold. Patient-bootstrap $\log_2$ expression intervals cross the accepted $[-0.5, 0.5]$ boundaries in every cohort, leaving the comparisons undecided. Depth matching discards roughly three quarters of the cells and pushes about half the remaining estimable rows below the count threshold. Within SMC, changing tissue eligibility changes the mature-cell pool from 662 to 3,824 cells. The across-cohort comparison in Section 3.6 instead holds tissue eligibility fixed. These observations restrict biological interpretation. Repeated cell draws and overlapping patient pairs do not supply independent patient replication.

C Supporting estimator and trial checks

Reference dependence. With nonempty stratum arms, saturated G-computation [Robins, 1986], inverse-probability weighting (IPW) using empirical stratum propensities, non-cross-fitted saturated augmented IPW (AIPW) and matching using every control all reproduce $T(D) = \sum_g (n_g/n)(\bar{Y}_{1g} - \bar{Y}_{0g})$. The IPW weights recover each stratum mean; the AIPW augmentation sums to zero within stratum. Ordinary least squares (OLS) with additive stratum indicators instead weights contrasts by normalized $n_g \hat{p}_g(1 - \hat{p}_g)$ [Angrist, 1998, Frisch and Waugh, 1933]. These known identities show why the reference determines what an exact recovery check establishes.

Learned-generator check. Seven generators comprise four Gaussian mixtures, a multilayer-perceptron (MLP) conditional Gaussian and two parametric controls. Training cohorts are simulated from a three-stratum model with confounded treatment assignment and a homogeneous additive effect of 3; learned generators are fitted to these cohorts, not to clinical data. Ten estimators are then applied to fresh synthetic cohorts from each fitted generator. Six sizes (100, 200, 500, 1000, 2000, 5000), six seed streams and 30 replicates schedule 75,600 evaluations on 7,560 cohorts; generator failures leave 75,580 evaluations on 7,558 cohorts. Of these, 74,726 estimate–drawn-reference pairs are finite. The 854 exclusions include 410 nonfinite references and 731 nonfinite estimates, overlapping in 287 evaluations.

Saturated G-computation has zero residual against $T(D)$ in all 42 generator–size settings because the equality holds on every evaluable synthetic dataset. It does not depend on learning the generator correctly. Nonsaturated ridge and MLP outcome fits can depart; the largest MLP residual under mixture generators is 3.3 outcome units. These checks concern the estimator–reference pair, not realism of learned patient trajectories.

D Synthetic external-control design

The fixed registry contains 50,000 simulated records with stratum shares (0.55, 0.30, 0.15), baseline outcomes $Y_0 \mid g \sim \mathcal{N}(\mu_g, 3^2)$, $\mu = (12, 8, 4)$, and effects $Y_1 - Y_0 = \delta_g r$, $\delta = (3, 1.5, 0.25)$, $r \sim \text{Gamma}(4, 0.25)$ (shape, scale). Responsiveness is fixed per record and hidden from estimators. Enrollment weights favor shares (0.15, 0.25, 0.60); both arms sample without replacement and are disjoint. The pooled target is a mixture of enrolled and control records, not the registry population itself. Normal outcomes can be negative: this is an abstract continuous-outcome experiment, not a fitted progression-free-survival model.

There are 200 study draws at each of $n = 50, 100, 200, 400, 800, 1600$, using one seed (20260915) and one fixed registry: 1,200 studies, eight estimators and five references yield 48,000 comparisons: 47,969 finite values and 31 exclusions from one study with an empty stratum arm at $n = 50$. The registry uses the same seed. The intended finite-cohort truth averages both potential outcomes over enrolled records; it is available to the simulator, not to estimators observing one outcome per record. Five case-mix shifts and 50 balance draws at $n = 1600$ provide controls. Threshold crossings in the saved shift table compare medians, not replicate-level decisions.

The nine prespecified falsification checks did not fire. They verify the constructed comparison, not clinical realism or the frequency of the validation mistake.

E The retained-gene control

The control-panel audit compares saved unmatched and depth-matched single-cell summaries with a separate bulk dataset. Where both mature fractions equal one (all epithelium, without marker-based subdivision), MS4A12 still changes by -0.967 to -1.000 across the tested cohort/matching strata; CDX2 on the same cells changes by -0.396 to -0.293 . Depth-matched MS4A12 changes are -0.971 to -0.960 . These observations weaken a uniform labeling or depth artifact explanation; they do not eliminate every measurement artifact. Across nondegenerate strata, the median baseline is 2.56 CP10K and tumor expression 0.104 CP10K. Strata classified as degenerate have a median of one relevant tumor cell and cannot establish a precise complete loss.

In GSE39582 (585 bulk samples), the saved MS4A12 $\log_2$ fold change is -8.18 . This corroborates tissue-level depletion on a separate platform; it does not identify a within-cell mechanism. Lower-abundance genes retain more signal in the single-cell summaries, arguing against a simple universal abundance-floor explanation, not proving absence of gene-specific dropout. Repository chronology places panel freezing before dataset registration; it cannot establish what evidence was known outside the repository. The panel's prior expectation of retention therefore needs reassessment.

F External audit and research context

Search and classification. A pre-registered queue contained 41 entries; 19 were screened out and seven classified within the 26 entries examined in order. An amendment before code inspection relaxed unsuccessful queries; the classification rule was unchanged. Generating-parameter references or quantities computed outside the estimator's observed records give NO. Same-record references require examination of inputs, weights, conditioning, nuisance fitting and observation process; YES requires affirmative evidence of functional identity. All seven recorded verdicts are NO: `caus1`, `target-trial-emulation`, `ecasim`, `PlasmodeSimulation`, `StatsPAI`, `diff-diff`, and `Auto-Empirical-Research-Skills`. The first six use generating parameters or outside-sample quantities; the last compares cross-fitted partialling-out with full-sample OLS. Its 4.0×10^{-14} residual on noiseless linear data becomes 0.125 after adding outcome noise. The functions agree on the restricted noiseless support but not on general noisy datasets; the NO verdict concerns shared functional form. Classification used one reader, only this last repository was executed, and the sample is nonrandom. The virtual-control-arm query located no examinable code. These limits preclude a prevalence estimate.

Related scope. Cell-count filters are established in pseudobulk analysis [Crowell et al., 2020]; `scCODA`, `Milo`, `propeller` and `DA-seq` address composition [Büttner et al., 2021, Dann et al., 2022, Phipson et al., 2022, Zhao et al., 2021], while `cocoa` separates composition and expression changes [Petukhov et al., 2022]. Selective prediction motivates evaluating both answers and abstentions [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. Conformal prediction coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] does not define a missing conditional population or automatically provide intervals for this intrinsic parameter.

Documentary source of the case. The anonymous review record includes the original validation statement (Artifact A), interval criteria (Artifact B), and numerical context. Exact source mapping and checksums remain outside the review bundle. This records one internal validation error; the external search found no positive case.

Implementation and provenance. Corrections separated collapsed population definitions, aligned labeling, tracked producing scripts, and repaired undefined-interval denominators. Saved tables retain commit hashes and seeds; these establish traceability, not statistical validity.

Use of large language models. Language-model assistants supported coding, figures, drafting and model audits under author direction. They helped identify defects and introduced some of those above. Authors remain responsible for all claims. Automated checks re-derive selected results and complement review; they do not verify every manuscript statement.